Calibration

Initial setup and later recals. A full first run does every stage. A later run can open one process. Header prefix is `CAL ·`.

Draft. Pixel layout: UI/calibration.html. Hardware procedures: Limit_Marker_And_Snug.md, Load_Bin_Obstruction_Calibration.md.

The door will move. Path-clear must pass before any stage that commands the motor. There is no physical E-stop. HOLD cancels: PWM off, wait for current decay, leave K1 unchanged, discard the partial result, return to the stage list. Never write NVS mid-motion. Review is the single save.

1. Stage list

Entry for a full setup and for a single recalibration. Status words name stale or missing work so you do not run a later stage on a bad earlier one. Six compact rows fit. TURN windows the list. All eight stages are on this shot.

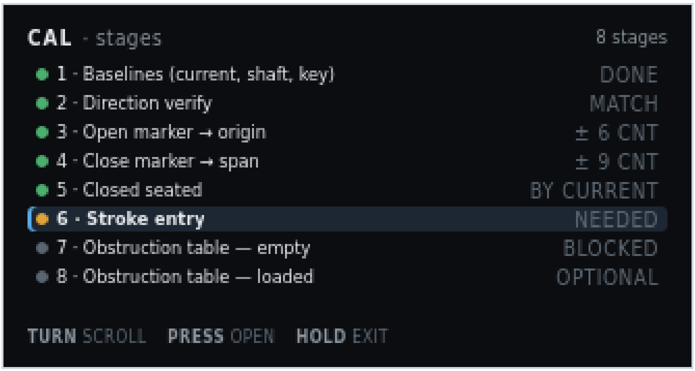

Figure 1. CAL · stages. Stroke entry selected. Stages 7–8 blocked until stroke exists.

On screen	Meaning
`CAL · stages`	Header left. Calibration mode, stage list.
`8 stages`	Header right. Count of guided processes. Review is a ninth screen after these, not a stage.
Green pip + `1 · Baselines (current, shaft, key)` + `DONE`	Stage 1 complete. Three rest captures: ACS712 zero, AS5600 jitter, 49E levels.
Green pip + `2 · Direction verify` + `MATCH`	Commanded direction agreed with the shaft. Invert flag is good. Gates all seeks.
Green pip + `3 · Open marker → origin` + `± 6 CNT`	Open-edge probes accepted. Spread across repeats is 6 counts. That edge is count 0.
Green pip + `4 · Close marker → span` + `± 9 CNT`	Close-edge probes accepted. Spread 9 counts. Edge minus origin is the measured span.
Green pip + `5 · Closed seated` + `BY CURRENT`	Seated found by current rise, not stall. Confirmation, not a home.
Amber pip + highlight + `6 · Stroke entry` + `NEEDED`	Selected row. Operator inches are missing. Required before learn runs, not before marker seeks.
Grey pip + `7 · Obstruction table — empty` + `BLOCKED`	Cannot start until stroke exists. Empty-rack learn.
Grey pip + `8 · Obstruction table — loaded` + `OPTIONAL`	Second table with racks at max expected load. Skip is allowed.
`TURN scroll`	Move the highlight. Window the list if more rows exist than fit.
`PRESS open`	Enter the highlighted stage. Moving stages show path-clear first.
`HOLD exit`	Return to Main. Incomplete RAM results stay until Review saves or you discard them there.

2. Path-clear gate

Shown before any stage that commands the motor. A checklist, not a single OK. The door does not move until every line is ticked.

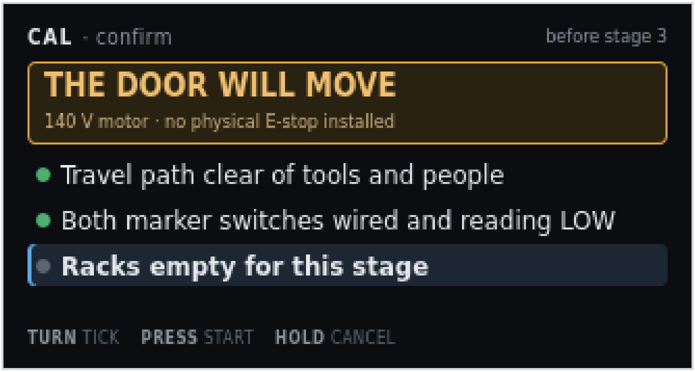

Figure 2. CAL · confirm, before stage 3. Third line still unticked.

On screen	Meaning
`CAL · confirm`	Header left. Gate, not a capture screen.
`before stage 3`	Header right. This shot is the gate for open-marker seek. Other moving stages show their own number.
`THE DOOR WILL MOVE`	Amber banner title. Condition before command.
`140 V motor · no physical E-stop installed`	Banner detail. HOLD is the only cancel. There is no E-stop on this build.
Green pip + `Travel path clear of tools and people`	Ticked. Operator has acknowledged a clear path.
Green pip + `Both marker switches wired and reading LOW`	Ticked. NC markers at rest read LOW on this polarity. A HIGH rest would mean a cut wire or a made switch.
Grey pip + highlight + `Racks empty for this stage`	Not yet ticked. Selected line. TURN moves among lines. This line appears when the stage requires empty racks.
`TURN tick`	Toggle the highlighted checklist line.
`PRESS start`	Start the stage when every line is ticked. Does nothing while a line is still grey.
`HOLD cancel`	Return to the stage list. No motion.

3. Stage 1a — current zero

Captured with PWM at zero. ACS712 offset only. It shares nothing with the shaft or key baselines.

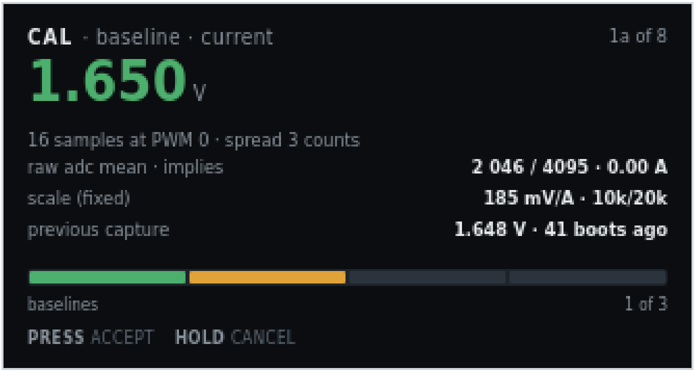

Figure 3. CAL · baseline · current. Motor at rest.

On screen	Meaning
CAL · baseline · current	Header left. First of three baseline captures.
1a of 8	Header right. Stage 1, first sensor, of eight stages.
1.650 V (green)	Hero. Mean ACS712 output at PWM 0. Green means the spread is acceptable.
16 samples at PWM 0 · spread 3 counts	How the mean was taken. Spread is ADC counts, not amps.
raw adc mean · implies · 2 046 / 4095 · 0.00 A	Raw mean of 4095, and the cooked amp reading that offset implies (zero at rest).
scale (fixed) · 185 mV/A · 10k/20k	ACS712-05B sensitivity and the Board A divider. Not an operator edit.
previous capture · 1.648 V · 41 boots ago	Last stored zero, for comparison. No RTC. Age is boot count.
Progress bar, first segment amber, baselines · 1 of 3	Three baseline screens. This is the first. Green segments are done. Amber is now. Grey is remaining.
PRESS accept	Keep this zero in RAM and go to shaft jitter.
HOLD cancel	Discard this capture. Return to the stage list.

4. Stage 1b — shaft jitter band

Measured at rest. This band sets stall minimum progress and the accumulation deadband.

Figure 4. CAL · baseline · shaft. Rest jitter over 2.0 s.

On screen	Meaning
`CAL · baseline · shaft`	Header left. Second baseline.
`1b of 8`	Header right.
`± 3 cnt` (green)	Hero. Peak-to-peak AS5600 counts at rest.
`peak-to-peak at rest · 2.0 s window`	How long the band was watched.
`samples · magnet` · `I2C ACK · detected`	Bus then STATUS. `I2C ACK` means the chip answered. `detected` / `lost` / `too weak` / `too strong` is magnet quality. `I2C NACK` plus `unread` means the bus failed.
`sets stall minimum progress` · `8 cnt / 500 ms`	Stall check needs more motion than rest jitter. Derived from this band, not guessed.
`sets accumulation deadband` · `3 cnt`	Counts inside this band do not walk idle position.
Progress, two green + one amber, `baselines` · `2 of 3`	Current zero accepted. Shaft is now. Key remains.
`PRESS accept` · `HOLD cancel`	Keep the band, or discard and return to the list.

5. Stage 1c — bottle key levels

Two captures: bottle out, then bottle in. Enter and exit thresholds are derived from the gap.

Figure 5. CAL · baseline · key. Second capture, bottle in the recess.

On screen	Meaning
`CAL · baseline · key`	Header left. Third baseline.
`1c of 8`	Header right.
`2 980` (green)	Hero. Present-level ADC this capture.
`capture 2 of 2 — bottle IN the recess`	Operator has placed the bottle. Capture 1 was absent.
`absent 1 605`	Right of the hero. First capture, bottle out.
`captured 00:41 ago`	Age of the absent capture this session.
`gap between levels · 1 375 counts`	Present minus absent. A small gap means a weak magnet or a bad placement.
`derived enter / exit · dead zone · 2 400 / 2 100 · 300`	Hysteresis pair and the dead zone between them. Not a single threshold.
`polarity for this build · absent → OPEN`	Default handbook polarity. Place the bottle to close. Change it in Settings if this cabinet is wired the other way.
Progress, three green + one amber, `baselines · 3 of 3`	Last baseline. Amber segment is this capture.
`PRESS accept · HOLD cancel`	Keep both levels, or discard.

6. Stage 2 — direction verify

Runs before any seek. A wrong invert flag would send the first seek at the far end of travel. Slow duty, short move. The operator confirms the door went the way the screen says.

Figure 6. CAL · direction. Match. Invert is unchanged.

On screen	Meaning
`CAL · direction`	Header left.
`2 of 8`	Header right.
`MATCH` (green)	Hero. Commanded OPEN and the shaft turned the expected way.
`commanded OPEN · shaft turned CCW`	What firmware asked and what the encoder saw.
`+412 cnt`	Net counts during the short move.
`duty 96 · 1.4 s`	Creep duty and how long the probe ran.
`invert flag` · `false (unchanged)`	Mapping of K1 to shaft sense. False means the compiled default already matches this cabinet.
`current during move` · `0.62 A peak`	Peak amps on this short move. A stall here is a wiring or bind problem, not a direction problem.
`door moved toward` · `OPEN — confirm on the door`	Look at the physical door. The encoder can match a wrong invert if you never look.
Amber line: `If the door moved to CLOSE, press to set the invert flag and repeat.`	Mismatch path. PRESS sets invert and runs the probe again.
`PRESS invert`	Flip the invert flag and repeat the short move.
`TURN accept`	Keep MATCH (or the new invert after a repeat).
`HOLD cancel`	Discard. Return to the stage list.

7. Stage 3 — open marker and origin

Repeat probes of the same edge from the same direction. The spread is the pass criterion. The accepted actuation edge becomes count zero.

Figure 7. CAL · open marker. Third probe complete. Spread 6 counts.

On screen	Meaning
`CAL · open marker`	Header left.
`3 of 8`	Header right.
`PROBE 3 / 3` (amber)	Hero. Last of three approaches. Amber because the motor just moved.
`creep · back off 400 cnt · re-approach`	Probe method. Operational snug does not apply during calibration seeks.
`probe 1 — actuate / release` · `402 / 361 cnt`	First pair. Actuation and release differ because the switch has hysteresis. Approach from the same direction each time.
`probe 2 — actuate / release` · `398 / 358 cnt`	Second pair.
`probe 3 — actuate / release` · `404 / 363 cnt`	Third pair.
`spread · tolerance` · `6 · 20 cnt`	Actuate spread across repeats versus the allowed band. 6 is inside 20.
`origin will be set at` · `actuate edge → 0`	Accepted open actuation becomes count zero for everything downstream.
`PRESS accept origin`	Keep the origin in RAM.
`HOLD cancel`	Discard a half-probed edge. Partial marker data is not usable.

8. Stage 4 — close marker and span

Same probe method at the other end. The accepted edge minus origin is the measured span. Obstruction bins divide this span.

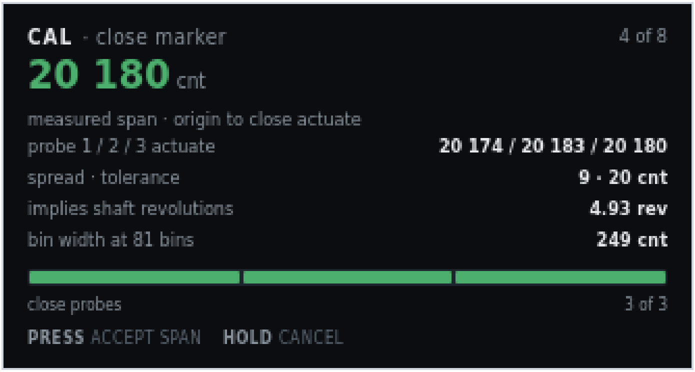

Figure 8. CAL · close marker. Measured span accepted if you press.

On screen	Meaning
`CAL · close marker`	Header left.
`4 of 8`	Header right.
`20 180 cnt` (green)	Hero. Measured span from origin to close actuation.
`measured span · origin to close actuate`	What the hero number is. Not seated. Seated is stage 5, past this edge.
`probe 1 / 2 / 3 actuate` · `20 174 / 20 183 / 20 180`	Three actuation edges. Release edges were captured. This row shows actuate only.
`spread · tolerance` · `9 · 20 cnt`	Spread inside tolerance.
`implies shaft revolutions` · `4.93 rev`	Span divided by 4096 counts/rev. Sanity check against the mechanical stroke.
`bin width at 81 bins` · `249 cnt`	Span divided into 81 obstruction bins.
Progress, three green, `close probes` · `3 of 3`	All three close probes done.
`PRESS accept span` · `HOLD cancel`	Keep the span, or discard the half-probed edge.

9. Stage 5 — closed seated

Past the close marker on creep. How the stop was found is stored: a current rise means the seal is loading. A stall without one means the door hit something solid.

Figure 9. CAL · closed seated. Found by current rise, 168 counts past the marker.

On screen	Meaning
`CAL · closed seated`	Header left.
`5 of 8`	Header right.
`SEATED` (green)	Hero. Physical closed stop found.
`found by CURRENT RISE at 1.90 A`	Method. Prefer current-draw rise (snug). Else record stall.
`+168 cnt`	Counts past the close-marker actuation.
`past close marker`	Seated is beyond the near-end flag, not at it.
`method stored` · `current rise (not stall)`	Written with the capture so Diagnostics can show why seated was accepted.
`creep duty · timeout` · `96 · 1.4 s of 2.0 s`	Duty used and how much of the seek timeout was consumed.
`overtravel cap` · `220 cnt — not reached`	Hard cap past the marker. A hit here is a fault, not a seated capture.
`seated position` · `20 348 cnt`	Origin-relative counts of the seated stop.
Green line: `Seated is a confirmation, not a position reference. Re-homing uses the marker edges.`	Do not treat seated as a home. Power-loss recovery seeks the open marker.
`PRESS accept` · `HOLD cancel`	Keep seated, or discard.

10. Stage 6 — stroke entry

The only operator-typed number, and the only place inches enter the system. It gates learn runs because bin widths are reported in centimetres. It never widens a safety limit.

Figure 10. CAL · stroke. 32.0 in, derived 631 cnt/in.

On screen	Meaning
`CAL · stroke`	Header left.
`6 of 8`	Header right.
`32.0` with caret	Pending inches, open marker to closed seated. TURN adjusts.
`inches, open marker → closed seated`	What the number measures. Not overall cabinet width.
`derived 631 cnt/in from 20 348 cnt`	Seated counts divided by the typed inches. Display only.
`geometric estimate · 640 cnt/in (unverified)`	Independent estimate from the mechanical drawing. Unverified means it is not a measured fact.
`difference · 1.4% — plausible`	Typed stroke versus the geometric estimate. A large disagreement warns. It does not block.
`used for · display and warnings only`	Inches and counts-per-inch never feed a seek cap.
`never used for · seek caps, snug window, bins`	Those limits stay in counts from stages 3–5.
Green line: `Warns on a large disagreement. It does not block, and it can be re-entered later.`	You can fix a typo without repeating marker probes.
`TURN adjust`	Change the pending inches.
`PRESS accept`	Keep the stroke in RAM. Unblocks stage 7.
`HOLD cancel`	Leave the previous stroke (or none).

11. Stage 7 — obstruction table, empty racks

Three clear runs each way at the production profile. Obstruction reverse is off. The 2.50 A ceiling stays on. A trip discards that run instead of teaching it.

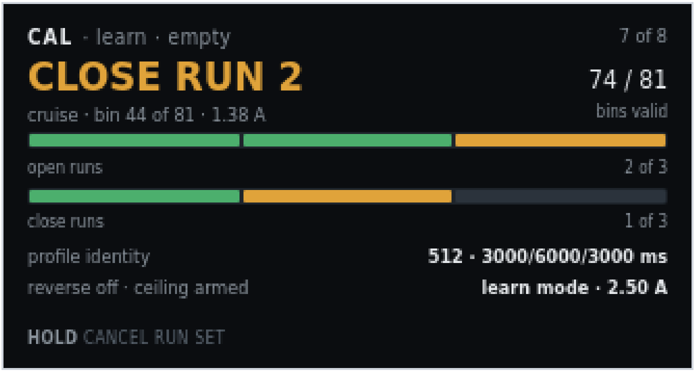

Figure 11. CAL · learn · empty. Second close run in cruise.

On screen	Meaning
`CAL · learn · empty`	Header left. Empty-rack table.
`7 of 8`	Header right.
`CLOSE RUN 2` (amber)	Hero. Second of three close strokes. Motor live.
`cruise · bin 44 of 81 · 1.38 A`	Profile phase, current bin, live amps.
`74 / 81`	Bins that already have a valid sample this run set.
`bins valid`	Learn runs stop at the marker, so some end bins stay unvisited.
Open-run pips, two green + one amber, `open runs` · `2 of 3`	Open-direction progress. This shot is mid close-run, so the open bar may already be ahead.
Close-run pips, one green + one amber + one grey, `close runs` · `1 of 3`	Close-direction progress. Run 2 is live.
`profile identity` · `512 · 3000/6000/3000 ms`	Cruise duty and accel / cruise / decel times. Stored with the table so a later profile change can invalidate it.
`reverse off · ceiling armed` · `learn mode · 2.50 A`	Obstruction reverse is off so a bind is not taught as load. The hard ceiling still trips and discards the run.
`HOLD cancel run set`	PWM off, current decay, K1 unchanged, discard the incomplete table. Return to the stage list. There is no PRESS on this live run.

12. Stage 8 — obstruction table, loaded racks

Optional second table with racks at maximum expected load. Which table arms in service is still an open spec decision, so the screen states it.

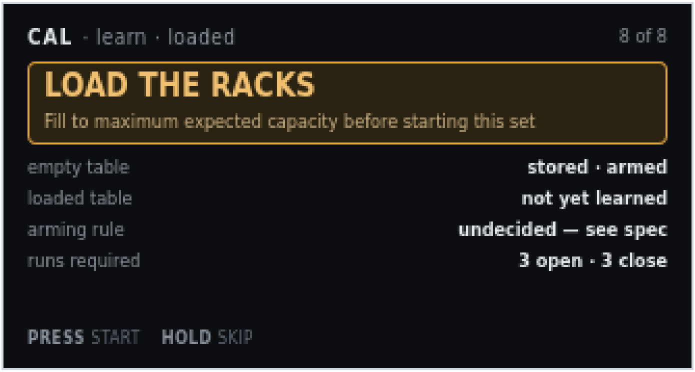

Figure 12. CAL · learn · loaded. Gate before the loaded run set.

On screen	Meaning
`CAL · learn · loaded`	Header left.
`8 of 8`	Header right.
`LOAD THE RACKS`	Amber banner title. Do not start this set empty.
`Fill to maximum expected capacity before starting this set`	Banner detail. Max expected load, not an overload test.
`empty table` · `stored · armed`	Stage 7 already produced a usable empty-rack envelope.
`loaded table` · `not yet learned`	This stage has not run.
`arming rule` · `undecided — see spec`	Which table Main uses in service is not chosen on this panel. See Production_Software.md.
`runs required` · `3 open · 3 close`	Same run count as the empty table.
`PRESS start`	After racks are loaded, start the run set (path-clear still applies).
`HOLD skip`	Leave the loaded table unlearned. Review will show it as optional skipped.

13. Review and save

Single NVS write. Everything above lived in RAM until this screen. The blob carries the profile identity that will later invalidate it. Motor must be idle.

Figure 13. CAL · review. One optional stage skipped. Motor idle.

On screen	Meaning
`CAL · review`	Header left.
`save`	Header right. This is the write point.
`READY TO SAVE` (green)	Hero. Completeness check passed.
`8 stages complete · 1 optional skipped`	Loaded-rack table was skipped. Empty table is enough to leave Needs Setup.
`origin · span · seated` · `0 · 20 180 · 20 348 cnt`	The three count references from stages 3–5.
`marker spread (open / close)` · `6 / 9 cnt`	Probe quality from stages 3 and 4.
`stroke → derived` · `32.0 in → 631 cnt/in`	Typed stroke and the display-only scale.
`tables · blob` · `empty armed · v3 3.6 kB`	Which envelope is armed, format version, and blob size.
Green line: `Motor is idle and current has decayed. NVS is never written while the motor runs.`	If the motor is still spinning, this screen will not write.
`PRESS save`	One NVS write. Then return toward Main.
`HOLD discard`	Drop the RAM results. Stored NVS is unchanged.

14. Knob on a live stage

Gesture	Typical meaning
TURN	Jog or adjust where the stage allows it (stroke entry, direction accept, checklist tick).
PRESS	Capture, accept, or start the next sample.
HOLD	Cancel the live procedure. Return to the stage list.

When an earlier value changes, mark dependent later steps stale. There is no RTC. Metadata uses run count and relative millis only. Never reverse K1 without PWM blanking.

15. Related

- Software overview — full web UI, figure 1

This page is a chapter of the Software User Manual.